

SAT Math Mastery · Week TP1J

SAT Trick Pack 1 — Part J

Time 120 minutes: 90 min solving + 30 min self-check and error log

Topics polygon angle sums · tangency · symmetry of a quadratic · unit chains · radical equations with a parameter · percent systems · fixed-plus-rate models · log vs line · proportional equations · difference of squares · grouping · exponent rewriting · conjugate factors · asymptotes

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Equate a sum of angles to $180(n-2)$, solve for n , and reject roots that break convexity.
- S2.** Turn tangency into a zero discriminant; find every parameter value and the contact points.
- S3.** Use equal outputs of a quadratic to place the axis at a midpoint, then match it to the zeros.
- S4.** Chain unit conversions in the right direction; square (or cube) the factor for area (or volume).
- S5.** Handle rates that change between segments and convert once at the end.
- S6.** Substitute $t = \sqrt{x}$ and count valid roots with $t \geq 0$.
- S7.** Write each category as a fraction of the original total and eliminate variables.
- S8.** Isolate the per-unit part of a fixed-plus-rate model before applying a percent change.
- S9.** Compare a line with a logarithmic curve by splitting the domain at the intersections.
- S10.** Make one linear equation a multiple of another and read off the scale factor.
- S11.** Spot $A^2 - B^2$ and common binomial factors before expanding.
- S12.** Rewrite radicals as powers of one base; read bounds and attainment of exponential decay.

2 · REQUIRED FACTS

- The interior angles of a convex n -gon sum to $180(n-2)$ and each is less than 180° .
- A line is tangent to a parabola when the combined quadratic has discriminant 0; the contact has $x = -\frac{b}{2a}$.
- $f(p) = f(q)$ for a quadratic puts the axis at $x = \frac{p+q}{2}$, which is also the midpoint of the zeros.
- To convert squared units, square the linear factor ($1 \text{ yd} = 3 \text{ ft} \Rightarrow 1 \text{ yd}^2 = 9 \text{ ft}^2$).
- $\sqrt{x} = t$ forces $t \geq 0$; a quadratic in t with negative product of roots has exactly one root $t > 0$.
- Two equations $ax+by=c$ and $a'x+b'y=c'$ have infinitely many solutions when (a', b', c') is a constant multiple of (a, b, c) .
- $A^2 - B^2 = (A-B)(A+B)$; $a^m \cdot a^n = a^{m+n}$; $\sqrt[n]{a^m} = a^{m/n}$.
- A function $c + dr^x$ with $0 < r < 1$ and $d > 0$ decreases toward c but never reaches it.

3 · REQUIRED FORMULAS

$$\sum \theta = 180(n-2) = \frac{n}{2}(2a + (n-1)d), \quad \Delta = b^2 - 4ac = 0, \quad x_{\text{axis}} = \frac{p+q}{2}$$

$$A^2 - B^2 = (A-B)(A+B), \quad (u+v)(u-v) = u^2 - v^2, \quad \text{fare} = \text{fee} + \text{rate} \cdot d$$

4 · MINI THEORY SHEET

Parameter conditions. Translate the wording into one equation: sum of angles = $180(n-2)$, discriminant = 0, axis = midpoint, scale factor equal. Solve it completely, then test every root against the stated restrictions (convexity, $t \geq 0$, domain).

Units. Write the conversion as a fraction equal to 1 and make the unwanted unit cancel. For area use the squared factor. Convert at the end, not midway.

Expressions. Before expanding, look for a difference of squares or a shared binomial; substituting the given relation into the factored form is shorter and safer.

Functions. For a curve against a line, find the intersections, then test one point in every interval. For decay terms ask three separate questions: where does it head, is it ever reached, which way does it move?

Speed shortcuts.

Situation	Shortcut
angles in an arithmetic sequence	series sum = $180(n - 2)$; reject a root giving an angle $\geq 180^\circ$
line tangent to a parabola	discriminant = 0; contact at $x = -\frac{b}{2a}$; count both parameter values
equal outputs, zeros given	axis = midpoint of the inputs = midpoint of the zeros
mixed units in a trip	total each unit separately, convert once
map scale and area	convert each side, then square the factor for the area unit
$x - c\sqrt{x} = k$	$t = \sqrt{x}$; one valid root if the constant term is negative or the root is double
percent changes by category	fractions of the original total; subtract the total equation
fee plus rate, then rate changes	subtract the two data points for the rate; change only the rate
line against log curve	intersections first, then test a point in each region
infinitely many solutions	divide matching coefficients to get the multiple; apply it to the constants
$(3a + b)^2 - (a + 3b)^2$	factor as $(A - B)(A + B)$, then use the given relation
four-term expression and a given sum	group to expose the given binomial
roots and powers of one variable	everything to one base, then add exponents
$(Ax + B - \sqrt{c})(Ax + B + \sqrt{c})$	$= (Ax + B)^2 - c$; leading factor cancels in ratios
$c + dr^x, 0 < r < 1$	bound c is approached, never attained; function is monotone

5 · GUIDED EXAMPLES

Example 1 — Tangency: zero discriminant, both parameter values

The line $y = kx - 1$ is tangent to the parabola $y = x^2 + 2x + 3$. Find every possible k .

Set equal: $x^2 + (2 - k)x + 4 = 0$. Tangent means $\Delta = (2 - k)^2 - 16 = 0$, so $2 - k = \pm 4$ and $k = -2$ or $k = 6$. Both work.

Example 2 — Difference of squares before expanding

If $p^2 - q^2 = 15$, find $(2p + q)^2 - (p + 2q)^2$.

$(2p + q - p - 2q)(2p + q + p + 2q) = (p - q) \cdot 3(p + q) = 3(p^2 - q^2) = 45$.

Example 3 — Map scale, then square the factor

On a map, 1 cm represents 200 m. A lake is a 4 cm by 3 cm rectangle on the map. Find its real area in hectares (1 ha = 10,000 m²).

Real sides: 800 m and 600 m, area 480,000 m² = 48 ha. (Shortcut: $12 \text{ cm}^2 \times 200^2 = 480,000 \text{ m}^2$.)

6 · PRACTICE SET — 15 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

SAT CHALLENGE · NO CALCULATOR NEEDED

A convex polygon has n sides. Its interior angles, listed from smallest to largest, form an arithmetic sequence with smallest angle 120° and common difference 5° . How many sides does the polygon have?

(A) 7

(B) 9 ✓

(C) 16

(D) 18

1. Sum of the angles as an arithmetic series: $\frac{n}{2}(2 \cdot 120 + (n-1) \cdot 5) = \frac{n}{2}(235 + 5n)$.

2. Angle sum of an n -gon: $180(n-2)$. Equate: $n(235 + 5n) = 360(n-2) \Rightarrow 5n^2 - 125n + 720 = 0 \Rightarrow n^2 - 25n + 144 = 0$.

3. Factor: $(n-9)(n-16) = 0$, so $n = 9$ or $n = 16$.

4. Convexity needs every angle below 180° . Largest angle: $120 + 5(n-1)$. For $n = 16$ it is $195^\circ > 180^\circ$, so reject.

5. For $n = 9$ the largest angle is $160^\circ < 180^\circ$ and the sum is $\frac{9}{2}(240 + 40) = 1260 = 180 \cdot 7$. The polygon has 9 sides.

Answer: (B)

2.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the line $y = kx + 4$, where k is a constant, is tangent to the parabola $y = x^2 + 4x + 13$ for exactly two values of k . For each of these values the line touches the parabola at one point. What is the greatest y -coordinate of these two points of tangency?

(A) 10

(B) 13

(C) 34 ✓

(D) 3

1. Set the expressions equal: $x^2 + 4x + 13 = kx + 4 \Rightarrow x^2 + (4-k)x + 9 = 0$.

2. Tangent means exactly one solution, so the discriminant is 0: $(4-k)^2 - 36 = 0 \Rightarrow 4-k = \pm 6 \Rightarrow k = -2$ or $k = 10$.

3. The double root is $x = -\frac{4-k}{2}$: for $k = 10$, $x = 3$; for $k = -2$, $x = -3$.

4. y -coordinates (use the parabola): $y(3) = 9 + 12 + 13 = 34$ and $y(-3) = 9 - 12 + 13 = 10$. Check with the line: $10(3) + 4 = 34$ and $-2(-3) + 4 = 10$.

5. The greater y -coordinate is 34.

Answer: (C)

3.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function $g(x) = 2x^2 + bx + c$, where b and c are constants, has zeros $k + 1$ and $3k - 5$, where k is a constant. If $g(2) = g(10)$, what is the value of c ?

(A) 70 ✓

(B) 140

(C) 24

(D) 35

1. $g(2) = g(10)$ puts the axis of symmetry at $x = \frac{2+10}{2} = 6$.

2. The axis is also the midpoint of the zeros: $\frac{(k+1) + (3k-5)}{2} = 2k - 2 = 6 \Rightarrow k = 4$. The zeros are 5 and 7.

3. $g(x) = 2(x-5)(x-7) = 2x^2 - 24x + 70$, so $c = 70$.

4. Check: $g(2) = 8 - 48 + 70 = 30$ and $g(10) = 200 - 240 + 70 = 30$.

Answer: (A)

4.

SAT CHALLENGE · CALCULATOR

A car uses fuel at a rate of 6.0 liters per 100 kilometers on a highway and 10.0 liters per 100 kilometers in the city. On one trip the car travels 240 kilometers on the highway and 80 kilometers in the city. Using 1 mile = 1.6 kilometers and 1 gallon = 3.8 liters, which of the following is closest to the car's fuel efficiency for the whole trip, in miles per gallon?

(A) 87

(B) 6

(C) 30

(D) 34 ✓

1. Highway fuel: $\frac{240}{100} \cdot 6.0 = 14.4$ L. City fuel: $\frac{80}{100} \cdot 10.0 = 8.0$ L. Total fuel: 22.4 L.

2. Total distance: $240 + 80 = 320$ km.

3. Convert once at the end: distance = $\frac{320}{1.6} = 200$ miles; fuel = $\frac{22.4}{3.8} \approx 5.895$ gallons.

4. Efficiency = $\frac{200}{5.895} \approx 33.9$ miles per gallon, closest to 34.

5. (Averaging the two rates gives 8 L/100 km and about 30 mpg, which ignores that more distance is on the highway.)

Answer: (D)

5.

SAT CHALLENGE · NO CALCULATOR NEEDED

On a map, 1 inch represents 1,500 feet. A rectangular parking lot appears on the map as 2.4 inches by 1.6 inches. Which of the following is the actual area of the lot, in square yards? (1 yard = 3 feet)

(A) 8,640,000

(B) 320,000

(C) 960,000 ✓

(D) 2,880,000

1. Actual sides: $2.4 \cdot 1,500 = 3,600$ ft and $1.6 \cdot 1,500 = 2,400$ ft.

2. Convert each side to yards: $\frac{3600}{3} = 1200$ yd and $\frac{2400}{3} = 800$ yd.

3. Area = $1200 \cdot 800 = 960,000$ square yards.

4. (Equivalent: $8,640,000 \text{ ft}^2 \div 9 = 960,000 \text{ yd}^2$, since $1 \text{ yd}^2 = 9 \text{ ft}^2$. Dividing by 3 or 27 gives the other options.)

Answer: (C)

6.

SAT CHALLENGE · NO CALCULATOR NEEDED

For a real number k , consider the equation

$$x - 4\sqrt{x} = k.$$

For how many integer values of k from -10 to 10 , inclusive, does the equation have exactly one real solution x ?

1. Let $t = \sqrt{x}$ with $t \geq 0$. The equation becomes $t^2 - 4t - k = 0$; each root $t \geq 0$ gives exactly one $x = t^2$.

2. Discriminant: $16 + 4k$. Real roots need $k \geq -4$. Roots: $t = 2 \pm \sqrt{4 + k}$.

3. Case $k = -4$: double root $t = 2$, one solution ($x = 4$).

4. Case $k > 0$: product of roots = $-k < 0$, so one root is negative (rejected) and one is positive: exactly one solution.

5. Case $-4 < k \leq 0$: product $-k \geq 0$ and sum $4 > 0$, so both roots are nonnegative and distinct: two solutions. Case $k < -4$: none.

6. Valid integers: $k = -4$ and $k = 1, 2, \dots, 10$, which is $1 + 10 = 11$.

Answer: 11

7.

SAT CHALLENGE · NO CALCULATOR NEEDED

A library's collection consists of fiction, nonfiction and reference books. Next year the number of fiction books will increase by 20%, the number of nonfiction books will decrease by 10%, and the number of reference books will not change. After these changes, the collection will contain 1% more books than it does now, and the number of fiction books will equal the number of nonfiction books. What percent of the current collection is reference books?

(A) 35%

(B) 65% ✓

(C) 15%

(D) 20%

1. Let f , n , r be the fractions of the current collection that are fiction, nonfiction, reference: $f + n + r = 1$.

2. New total: $1.2f + 0.9n + r = 1.01$. Subtract the first equation: $0.2f - 0.1n = 0.01$.

3. Equal counts after the change: $1.2f = 0.9n \Rightarrow n = \frac{4}{3}f$.

4. Substitute: $0.2f - \frac{0.1 \cdot 4}{3}f = \frac{0.2}{3}f = 0.01 \Rightarrow f = 0.15$, so $n = 0.20$.

5. $r = 1 - 0.15 - 0.20 = 0.65$, i.e. 65%. (Check: $1.2(0.15) = 0.18 = 0.9(0.20)$ and $0.18 + 0.18 + 0.65 = 1.01$.)

Answer: (B)

8.

SAT CHALLENGE · CALCULATOR

A ride service charges a fixed pickup fee plus a constant rate for each mile traveled. A 5-mile ride costs \$14.50 and a 12-mile ride costs \$29.20. The service then raises its per-mile rate by 20% and leaves the pickup fee unchanged. Using 1 mile = 1.6 kilometers, what will a 20-kilometer ride cost after the change?

(A) \$36.30

(B) \$54.40

(C) \$84.64

(D) \$35.50 ✓

1. Subtract the two fares to remove the fixed fee: $\frac{29.20 - 14.50}{12 - 5} = \frac{14.70}{7} = 2.10$ dollars per mile.

2. Pickup fee: $14.50 - 5(2.10) = 4.00$ dollars.

3. New rate (only the rate rises): $1.20 \cdot 2.10 = 2.52$ dollars per mile.

4. Convert the distance: $20 \text{ km} = \frac{20}{1.6} = 12.5$ miles.

5. Cost = $4.00 + 2.52(12.5) = 4.00 + 31.50 = \35.50 .

6. (Raising the whole fare by 20% gives \$36.30; skipping the conversion gives \$54.40; multiplying by 1.6 instead of dividing gives \$84.64.)

Answer: (D)

9.

SAT CHALLENGE · DESMOS

Let $g(x) = \log_2 x$ for $x > 0$ and $h(x) = \frac{x+8}{6}$. For which values of x is $h(x) > g(x)$?

(A) $0 < x < 4$ or $x > 16$ ✓(B) $4 < x < 16$ (C) $x > 16$ only(D) $x < 4$ or $x > 16$

1. Look for intersections at powers of 2: $g(4) = 2 = h(4) = \frac{12}{6}$ and $g(16) = 4 = h(16) = \frac{24}{6}$.
2. The curve $y = \log_2 x$ is concave down, so a line can meet it in at most two points; these are the only two. They split $x > 0$ into three regions.
3. Test one value in each region. $x = 1$: $h = 1.5 > g = 0$. $x = 8$: $h = \frac{16}{6} \approx 2.67 < g = 3$. $x = 32$: $h \approx 6.67 > g = 5$.
4. So $h(x) > g(x)$ for $0 < x < 4$ and for $x > 16$.

Answer: (A)

10.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$3x + ky = 12$$

$$21x + 14y = 6m$$

In the system of equations above, k and m are constants. If the system has infinitely many solutions (x, y) , what is the value of km ?

1. Infinitely many solutions means the second equation is a constant multiple of the first. Matching the x -coefficients: $\frac{21}{3} = 7$.
2. y -coefficients: $7k = 14 \Rightarrow k = 2$.
3. Constants: $7 \cdot 12 = 6m \Rightarrow m = 14$. (The right side of the second equation is $6m$, not m .)
4. $km = 2 \cdot 14 = 28$. Check: $3x + 2y = 12$ times 7 gives $21x + 14y = 84 = 6(14)$.

Answer: 28

11.

SAT CHALLENGE · NO CALCULATOR NEEDED

For real numbers a and b , $a^2 - b^2 = 36$. What is the value of $(3a + b)^2 - (a + 3b)^2$?

1. Do not expand. Treat it as $A^2 - B^2$ with $A = 3a + b$ and $B = a + 3b$.
2. $A - B = 2a - 2b = 2(a - b)$ and $A + B = 4a + 4b = 4(a + b)$.
3. Product: $8(a - b)(a + b) = 8(a^2 - b^2) = 8 \cdot 36 = 288$.
4. (Expanding confirms it: $9a^2 + 6ab + b^2 - a^2 - 6ab - 9b^2 = 8a^2 - 8b^2$.)

Answer: 288**12.**

SAT CHALLENGE · NO CALCULATOR NEEDED

If $2x + 3y = 8$, which of the following is equivalent to $2x^2 + 3xy - 4x - 6y$?

(A) $8x - 4y$ **(B) $8x - 16$ ✓**(C) $8x - 8$ (D) $16 - 8x$

1. Group the terms: $(2x^2 + 3xy) - (4x + 6y) = x(2x + 3y) - 2(2x + 3y)$.
2. Factor out the common binomial: $(2x + 3y)(x - 2)$.
3. Substitute $2x + 3y = 8$: $8(x - 2) = 8x - 16$.
4. (Check with $x = 1$, $y = 2$, which satisfy $2x + 3y = 8$: the expression is $2 + 6 - 4 - 12 = -8$ and $8(1) - 16 = -8$.)

Answer: (B)

13.

SAT CHALLENGE · NO CALCULATOR NEEDED

Let t be a positive real number with $t \neq 1$. Define $m = \sqrt[3]{t^2}$ and $n = \sqrt{t}$. If

$$\frac{m^p \cdot n^3}{\sqrt[6]{t}} = t^5,$$

what is the value of p ?

1. Rewrite as powers of t : $m = t^{2/3}$, $n = t^{1/2}$, $\sqrt[6]{t} = t^{1/6}$.

2. Left side: $\frac{t^{2p/3} \cdot t^{3/2}}{t^{1/6}} = t^{2p/3+3/2-1/6} = t^{2p/3+4/3}$.

3. Since $t > 0$ and $t \neq 1$, equal powers have equal exponents: $\frac{2p}{3} + \frac{4}{3} = 5 \Rightarrow \frac{2p}{3} = \frac{11}{3} \Rightarrow p = \frac{11}{2}$.

4. Enter 11/2 or 5.5.

Answer: 11/2

14.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function $f(x) = 2(3x + 1 - \sqrt{5})(3x + 1 + \sqrt{5})$ can be written as $f(x) = ax^2 + bx + c$, where a , b and c are constants. What is the value of $\frac{b+c}{a}$?

1. The factors are conjugates of the form $(u-v)(u+v)$ with $u = 3x+1$ and $v = \sqrt{5}$, so the product is $u^2 - v^2 = (3x+1)^2 - 5 = 9x^2 + 6x - 4$.

2. Keep the leading factor 2: $f(x) = 2(9x^2 + 6x - 4) = 18x^2 + 12x - 8$. So $a = 18$, $b = 12$, $c = -8$.

3. $\frac{b+c}{a} = \frac{12-8}{18} = \frac{4}{18} = \frac{2}{9}$.

4. (Shortcut: the leading factor 2 multiplies every coefficient, so it cancels in any ratio: $\frac{b+c}{a} = \frac{6-4}{9} = \frac{2}{9}$.) Enter 2/9.

Answer: 2/9

15.

SAT CHALLENGE · NO CALCULATOR NEEDED

Let $g(x) = 7 + 9\left(\frac{1}{4}\right)^x$ for all real numbers x . Consider the following statements.

- I. $g(x) > 7$ for every real number x .
- II. The minimum value of g is 7.
- III. g is a decreasing function.

Which of the statements must be true?

(A) I and III only ✓

(B) I and II only

(C) II and III only

(D) I, II, and III

1. $\left(\frac{1}{4}\right)^x > 0$ for every x , so $9\left(\frac{1}{4}\right)^x > 0$ and $g(x) > 7$. Statement I is true.
2. As x grows, $\left(\frac{1}{4}\right)^x \rightarrow 0$, so $g(x) \rightarrow 7$, but the term is never 0, so $g(x) = 7$ is never attained. There is no minimum value; 7 is only a lower bound. Statement II is false.
3. The base $\frac{1}{4}$ is between 0 and 1, so $\left(\frac{1}{4}\right)^x$ decreases as x increases; multiplying by $9 > 0$ and adding 7 keeps that. Statement III is true.
4. Only I and III are true.

Answer: (A)